



MyHep ALL™

Sofosbuvir and Velpatasvir Film Coated Tablets

400 mg/100 mg

1. NAME OF THE MEDICINAL PRODUCT

MyHep ALL (Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg)

2. Qualitative and Quantitative Composition

Each Film Coated Tablet contains:

Sofosbuvir 400 mg

Velpatasvir 100 mg

Excipients with known effects: Each film-coated tablet contains 261.0 mg of lactose monohydrate.

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film coated tablet.

Light green to green colored capsule shaped biconvex beveled edge film coated tablet debossed with 'S' and P on either side of score line on one side and 'M' and V on other side of Score Line on another side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is indicated for the treatment of chronic hepatitis C virus (HCV) infection in adults (see sections Posology & Method of Administration, Warning & Precautions, and Pharmacodynamics).

4.2 Posology

The recommended dose of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is one tablet, taken orally, once daily with or without food (see section Pharmacokinetics).

Table 1: Recommended treatment and duration for all HCV genotypes

Patient population ^a	Treatment and duration
Patients with genotype 1, 2, 3, 4, 5 or 6 HCV	
Patients without cirrhosis and patients with compensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks
Patients with genotype 1 or 3 HCV	
Patients with decompensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + ribavirin ^b for 12 weeks

a. Includes patients co-infected with human immunodeficiency virus (HIV).

b. When administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg, the recommended dosage of ribavirin is based on weight (administered with food); 1000 mg per day for patients less than 75kg and 1200 mg for those weighing at least 75kg, divided and administered twice daily. The starting dosage and on-treatment dosage of ribavirin can be decreased based on haemoglobin and creatinine clearance. For ribavirin dosage modifications, refer to the package insert of medicinal products containing ribavirin.

Patients should be instructed that if vomiting occurs within 3 hours of dosing an additional tablet of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg should be taken. If vomiting occurs more than 3 hours after dosing, no further dose of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is needed (see section Pharmacodynamics).

If a dose of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is missed and it is within 18 hours of the normal time, patients should be instructed to take the tablet as soon as possible and then patients should take the next dose at the usual time. If it is after 18 hours then patients should be instructed to wait and take the next dose of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg at the usual time. Patients should be instructed not to take a double dose of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.

Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg treatment should be initiated and monitored by a physician experienced in the management of patients with HCV infection.

Elderly

No dose adjustment is warranted for elderly patients (see section Pharmacokinetics).

Renal impairment

No dose adjustment of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is required for patients with mild or moderate renal impairment.

Safety data are limited in patients with severe renal impairment (estimated glomerular filtration rate [eGFR] < 30 mL/min/1.73 m²) and end stage renal disease (ESRD) requiring haemodialysis. Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg can be used in these patients with no dose adjustment when no other relevant treatment options are available (see section Special warnings and precautions for use, Undesirable effects, Pharmacokinetics and Pharmacodynamics).

Hepatic impairment

No dose adjustment of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is required for patients with mild, moderate, or severe hepatic impairment (CPT Class A, B, or C) (see section Pharmacokinetics). Safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg have been assessed in patients with CPT Class B cirrhosis, but not in patients with CPT Class C cirrhosis (see sections Warning & Precautions, Undesirable Effects and Pharmacodynamics).

Pediatric population

The safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in children and adolescents aged less than 18 years have not yet been established. No data are available.

Method of administration

For oral use.

Patients should be instructed to swallow the tablet whole with or without food (see section Pharmacokinetics). Due to the bitter taste, it is recommended that the film-coated tablet is not chewed or crushed.

The score line is merely for aesthetic purposes and the tablet is not meant to cut into half

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section list of excipients.

Use with potent P-gp and potent CYP inducers

Medicinal products that are strong P-glycoprotein (P-gp) and/or strong cytochrome P450 (CYP) inducers (rifampicin, rifabutin, St. John's wort, carbamazepine, phenobarbital and phenytoin). Co-administration will significantly decrease sofosbuvir or velpatasvir plasma concentrations and could result in loss of efficacy of Sofosbuvir and Velpatasvir tablet (see section Interaction with other medicinal products and other forms of interaction).

4.4 Special warnings and precautions for use

Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg should not be administered concurrently with other medicinal products containing sofosbuvir.

Severe bradycardia and heart block

Cases of severe bradycardia and heart block have been observed when sofosbuvir-containing regimens are used in combination with amiodarone with or without other medicinal products that lower heart rate. The mechanism is not established.

The concomitant use of amiodarone was limited through the clinical development of sofosbuvir. Cases are potentially life threatening, therefore amiodarone should only be used in patients on Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg when other alternative anti-arrhythmic treatments are not tolerated or are contraindicated. Should concomitant use of amiodarone be considered necessary, it is recommended that patients are closely monitored when initiating Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. Patients who are identified as being at high risk of bradyarrhythmia should be continuously monitored for 46 hours in an appropriate clinical setting.

Due to the long half-life of amiodarone, appropriate monitoring should also be carried out for patients who have discontinued amiodarone within the past few months and are to be initiated on Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.

All patients receiving Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in combination with amiodarone with or without other medicinal products that lower heart rate should also be warned of the symptoms of bradycardia and heart block and should be advised to seek medical advice urgently should they experience them.

Patients who have previously failed therapy with an NS5A-containing regimen

There are no clinical data to support the efficacy of sofosbuvir/velpatasvir for the treatment of patients who have failed treatment with a regimen containing another NS5A inhibitor. However, on the basis of NS5A resistance associated variants (RAVs) typically seen in patients who have failed therapy with other NS5A inhibitor containing regimens, the *in vitro* pharmacology of velpatasvir, and the outcomes of sofosbuvir/velpatasvir treatment in NS5A-naïve patients with baseline NS5A RAVs enrolled into the ASTRA-studies, treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV for 24 weeks can be considered for patients who have failed therapy on an NS5A-containing regimen and who are deemed at high risk for clinical disease progression and who do not have alternative treatment options.

Renal impairment

Safety data are limited in patients with severe renal impairment (eGFR < 30 mL/min/1.73 m²) and ESRD requiring haemodialysis. Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg can be used in these patients with no dose adjustment when no other relevant treatment options are available (see sections Undesirable effects, Pharmacokinetics and Pharmacodynamics). When Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is used in combination with ribavirin refer also to the Package Insert for ribavirin for patients with creatinine clearance < 50 mL/min (see section Pharmacodynamics).

Use with moderate P-gp inducers or moderate CYP inducers

Medicinal products that are moderate P-gp or moderate CYP inducers (e.g. oxcarbazepine, modafinil or efavirenz) may decrease sofosbuvir or velpatasvir plasma concentrations leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. Co-administration of such medicinal products with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is not recommended (see section on interaction with other medicinal products and other forms of interaction).

Use with certain HIV antiretroviral regimens

Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg has been shown to increase tenofovir exposure, especially when used together with an HIV regimen containing tenofovir disoproxil fumarate and a pharmacokinetic enhancer (ritonavir or cobicistat). The safety of tenofovir disoproxil fumarate in the setting of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg and a pharmacokinetic enhancer has not been established. The potential risks and benefits associated with co-administration of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg with the fixed-dose combination tablet containing elvitegravir/cobicistat/tenofovir disoproxil fumarate or tenofovir disoproxil fumarate given in conjunction with a boosted HIV protease inhibitor (e.g. atazanavir or darunavir) should be considered, particularly in patients at increased risk of renal dysfunction. Patients receiving Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg concomitantly with elvitegravir/cobicistat/emtricitabine/tenofovir disoproxil fumarate or with tenofovir disoproxil fumarate and a boosted HIV protease inhibitor should be monitored for tenofovir-associated adverse reactions. Refer to tenofovir disoproxil fumarate, emtricitabine/tenofovir disoproxil fumarate, or elvitegravir/cobicistat/emtricitabine/tenofovir disoproxil fumarate Summary of Product Characteristics for recommendations on renal monitoring.

Use in diabetic patients

Diabetics may experience improved glucose control, potentially resulting in symptomatic hypoglycaemia, after initiating HCV direct-acting antiviral treatment. Glucose levels of diabetic patients initiating direct-acting antiviral therapy should be closely monitored, particularly within the first 3 months, and their diabetic medication modified when necessary. The physician in charge of the diabetic care of the patient should be informed when direct-acting antiviral therapy is initiated.

HCV/HBV (hepatitis B virus) co-infection

Cases of hepatitis B virus (HBV) reactivation, some of them fatal, have been reported during or after treatment with direct-acting antiviral agents. HBV screening should be performed in all patients before initiation of treatment. HBV/HCV co-infected patients are at risk of HBV reactivation, and should therefore be monitored and managed according to current clinical guidelines.

CPT Class C cirrhosis

Safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg has not been assessed in patients with CPT Class C cirrhosis (see sections Undesirable effects and Pharmacodynamic properties).

Liver transplant patients

The safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in the treatment of HCV infection in patients who are post-liver transplant have not been assessed. Treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in accordance with the recommended posology (see section Posology) should be guided by an assessment of the potential benefits and risks for the individual patient.

4.5 Interaction with other medicinal products and other forms of interaction

As Sofosbuvir and Velpatasvir tablet contains sofosbuvir and velpatasvir, any interactions that have been identified with these active substances individually may occur with Sofosbuvir and Velpatasvir tablet.

Potential for Sofosbuvir and Velpatasvir tablet to affect other medicinal products

Velpatasvir is an inhibitor of drug transporter P-gp, breast cancer resistance protein (BCRP), organic anion-transporting polypeptide (OATP) 1B1 and OATP1B3. Co-administration of Sofosbuvir and Velpatasvir tablet with medicinal products that are substrates of these transporters may increase the exposure of such medicinal products. See Table 3 for examples of interactions with sensitive substrates of P-gp (digoxin), BCRP (rosuvastatin), and OATP (pravastatin).

Potential for other medicinal products to affect Sofosbuvir and Velpatasvir tablet

Sofosbuvir and velpatasvir are substrates of drug transporters P-gp and BCRP. Velpatasvir is also a substrate of drug transporter OATP1B. *In vitro*, slow metabolic turnover of velpatasvir by CYP2B8, CYP2C8 and CYP3A4 was observed. Medicinal products that are potent inducers of P-gp or potent inducers of CYP2B8, CYP2C8, or CYP3A4 (e.g. rifampicin, rifabutin, St. John's wort, carbamazepine, phenobarbital and phenytoin) may decrease plasma concentrations of sofosbuvir or velpatasvir leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir. The use of such medicinal products with Sofosbuvir and Velpatasvir tablet is contraindicated (see section 4.3). Medicinal products that are moderate P-gp inducers or moderate CYP inducers (e.g. oxcarbazepine, modafinil, efavirenz or rifampine) may decrease sofosbuvir or velpatasvir plasma concentration leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir tablet. Co-administration with such medicinal products is not recommended with Sofosbuvir and Velpatasvir tablet (see section 4.4). Co-administration with medicinal products that inhibit P-gp or BCRP may increase sofosbuvir or velpatasvir plasma concentrations. Medicinal products that inhibit OATP CYP2B6, CYP2C8, or CYP3A4 may increase plasma concentration of velpatasvir. Clinically significant medicinal product interactions with Sofosbuvir and Velpatasvir tablet mediated by P-gp, BCRP, OATP or CYP450 inhibitors are not expected. Sofosbuvir and Velpatasvir tablet may be co-administered with P-gp, BCRP, OATP and CYP inhibitors.

Patients treated with vitamin K antagonists

As liver function may change during treatment with Sofosbuvir and Velpatasvir, a close monitoring of International Normalised Ratio (INR) values is recommended.

Impact of DAA therapy on drugs metabolized by the liver:

The pharmacokinetics of drugs that are metabolized by the liver (e.g. immunosuppressive agents such as calcineurin inhibitors) may be impacted by changes in liver function during DAA therapy, related to clearance of HCV.

Interactions between Sofosbuvir and Velpatasvir tablet and other medicinal products

Table 3 provides a listing of established or potentially clinically significant medicinal product interactions. The medicinal product interactions described are based on studies conducted with either Sofosbuvir and Velpatasvir or velpatasvir and sofosbuvir as individual agents, or are predicted medicinal product interactions that may occur with Sofosbuvir OR Velpatasvir. The table is not all-inclusive.

Table 3: Interactions between Sofosbuvir and Velpatasvir tablet and other medicinal products

Medicinal product by therapeutic area/Possible Mechanism of Interaction	Recommendation concerning co-administration with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg
ACID REDUCING AGENTS	
	Velpatasvir solubility decreases as pH increases. Medicinal products that increase gastric pH are expected to decrease the concentration of velpatasvir.
Antacids	
e.g. Aluminium or magnesium hydroxide; calcium carbonate	It is recommended to separate antacid and Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg administration by 4 hours.
H₂-receptor antagonists	
Famotidine (40 mg single dose)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose) ^a	H ₂ -receptor antagonists may be administered simultaneously with or staggered from Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg at a dose that does not exceed doses comparable to famotidine 40 mg twice daily
Famotidine dosed simultaneously with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a	
Cimetidine ^b	
Nizatidine ^b	
Ranitidine ^b	
(Increase in gastric pH)	
Famotidine (40 mg single dose)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose) ^a	
Famotidine dosed 12 hours prior to Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a	
(Increase in gastric pH)	
Proton pump inhibitors	
Omeprazole (20 mg once daily)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose fasted) ^a	Co-administration with proton pump inhibitors is not recommended. If it is considered necessary to co-administer, then Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg should be administered with food and taken 4 hours before proton pump inhibitor at max doses comparable to omeprazole 20 mg.
Omeprazole dosed simultaneously with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a	
Lansoprazole ^b	
Rabeprazole ^b	
Pantoprazole ^b	
Esomeprazole ^b	
(Increase in gastric pH)	
Omeprazole (20 mg once daily)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose fasted) ^a	
Omeprazole dosed 4 hours after Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a	
(Increase in gastric pH)	
ANTIARRHYTHMICS	
Amiodarone	Use only if no other alternative is available. Close monitoring is recommended if this medicinal product is administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg (see sections 4.4 and 4.8).
Digoxin	Co-administration of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg with digoxin may increase the concentration of digoxin. Caution is warranted and therapeutic concentration monitoring of digoxin is recommended when co-administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.
Digoxin (0.25 mg single dose)/ velpatasvir (100 mg single dose) (inhibition of P-gp)	
ANTICOAGULANTS	
Dabigatran etexilate (inhibition of P-gp)	Clinical monitoring, looking for signs of bleeding and anaemia, is recommended when dabigatran etexilate is co-administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. A coagulation test helps to identify patients with an increased bleeding risk due to increased dabigatran exposure.
Vitamin K antagonists	Close monitoring of INR is recommended with all vitamin K antagonists. This is due to liver function changes during treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.
ANTICONSULSANTS	
Carbamazepine	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg is contraindicated with carbamazepine, phenobarbital and phenytoin, potent P-gp and CYP inducers (see section 4.3).
Phenytoin	
Phenobarbital (Induction of P-gp and CYPs)	
Oxcarbazepine (Induction of P-gp and CYPs)	Co-administration of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg with oxcarbazepine is expected to decrease the concentration of sofosbuvir and velpatasvir, leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. Co-administration is not recommended (see section 4.4).

4.6 Fertility, pregnancy and lactation

Pregnancy

There are no or limited amount of data from the use of sofosbuvir, velpatasvir or Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in pregnant women.

Sofosbuvir

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. It has not been possible to fully estimate exposure margins achieved for sofosbuvir in the rat relative to the exposure in humans at the recommended clinical dose.

Velpatasvir

Animal studies have shown a possible link to reproductive toxicity. As a precautionary measure, Sofosbuvir and Velpatasvir tablet use is not recommended during pregnancy.

Breast-feeding

It is unknown whether sofosbuvir, metabolites of sofosbuvir or velpatasvir are excreted in human milk. Available pharmacokinetic data in animals have shown excretion of velpatasvir and metabolites of sofosbuvir in milk. A risk to the newborns/infants cannot be excluded. Therefore, Sofosbuvir and Velpatasvir tablet should not be used during breast-feeding.

Fertility

No human data on the effect of Sofosbuvir and Velpatasvir tablet on fertility are available. Animal studies do not indicate harmful effects of sofosbuvir or velpatasvir on fertility. If ribavirin is co-administered with Sofosbuvir and Velpatasvir tablet, refer to the Summary of Product Characteristics for ribavirin for detailed recommendations regarding pregnancy, contraception, and breast-feeding.

4.7 Effects on ability to drive and use machines

Sofosbuvir and Velpatasvir tablet has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

The safety assessment of Sofosbuvir and Velpatasvir tablet was based on pooled Phase 3 clinical study data from patients with genotype 1, 2, 3, 4, 5 or 6 HCV infection (with or without compensated cirrhosis) including patients who received Sofosbuvir and Velpatasvir tablet for 12 weeks. In clinical studies, headache, fatigue and nausea were the most common treatment emergent adverse events reported. These and other adverse events were reported at a similar frequency in placebo treated patients.

Patients with decompensated cirrhosis

The safety profile of Sofosbuvir and Velpatasvir tablet has been evaluated in one open-label study involving patients with CPT Class B cirrhosis. The adverse events observed were consistent with expected clinical sequelae of decompensated liver disease, or the known toxicity profile of ribavirin for patients receiving Sofosbuvir and Velpatasvir tablet in combination with ribavirin. Patients with renal impairment

The safety of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg has been evaluated in a 12-week non-controlled study subjects with ESRD requiring dialysis. In this setting, exposure of sofosbuvir metabolite GS-331007 was 20-fold increased, exceeding levels where adverse reactions have been observed in preclinical trials. In this limited clinical safety data set, the rate of adverse events and deaths was not clearly elevated from what is expected in ESRD patients.

Description of selected adverse reactions

Cardiac arrhythmias

Cases of severe bradycardia and heart block have been observed when sofosbuvir containing regimens are used in combination with amiodarone and/or other medicinal products that lower heart rate (see sections 4.4 and 4.5).

Skin disorders

Frequency Not known: Stevens-Johnson syndrome

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdose

The highest documented doses of sofosbuvir and velpatasvir were a single dose of 1,200 mg and a single dose of 500 mg, respectively. In these healthy volunteer studies, there were no untoward effects observed at these doses levels, and adverse events were similar in frequency and severity to those reported in the placebo groups. The effects of higher doses/exposures are not known.

No specific antidote is available for overdose on Sofosbuvir and Velpatasvir tablet. If overdose occurs the patient must be monitored for evidence of toxicity. Treatment of overdose with Sofosbuvir and Velpatasvir tablet consists of general supportive measures including monitoring of vital signs, as well as observation of the clinical status of the patient. Haemodialysis can efficiently remove the predominant circulating metabolite of sofosbuvir, GS-331007, with an extraction ratio of 53%. Haemodialysis is unlikely to result in significant removal of velpatasvir, since velpatasvir is highly bound to plasma proteins.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Direct acting antiviral, ATC code: J05AP55

Mechanism of action

Sofosbuvir is a pan-genotypic inhibitor of the HCV NS5B RNA-dependent RNA polymerase, which is essential for viral replication. Sofosbuvir is a nucleotide prodrug that undergoes intracellular metabolism to form the pharmacologically active uridine analogue triphosphate (GS-461203), which can be incorporated into HCV RNA by the NS5B polymerase and acts as a chain terminator. GS-461203 (the active metabolite of sofosbuvir) is neither an inhibitor of human DNA and RNA polymerases nor an inhibitor of mitochondrial RNA polymerase. Velpatasvir is a HCV inhibitor targeting the HCV NS5A protein, which is essential for both RNA replication and the assembly of HCV virions. *In vitro* resistance selection and cross-resistance studies indicate velpatasvir targets NS5A as its mode of action.

Antiviral activity

The 50% effective concentration (EC₅₀) values of sofosbuvir and velpatasvir against full-length or chimeric replicons encoding NS5B and NS5A sequences from the laboratory strains are presented in Table 3. The EC₅₀ values of sofosbuvir and velpatasvir against clinical isolates are presented in Table 4.

Table 3: Activity of sofosbuvir and velpatasvir against full-length or chimeric laboratory replicons

Replicon genotype	Sofosbuvir EC ₅₀ , nM ^a	Velpatasvir EC ₅₀ , nM ^b
1 ^a	40	0.014
1 ^b	110	0.016
2 ^a	50	0.005-0.016 ^c
2 ^b	15 ^c	0.002-0.006 ^c
3 ^a	50	0.004
4 ^a	40	0.009
4 ^b	NA	0.004
5 ^a	15 ^a	0.021-0.054 ^d
6 ^a	14 ^a	0.006-0.009
6 ^b	NA	0.130 ^e

NA = Not available

a. Mean value from multiple experiments of same laboratory replicon.

b. Stable chimeric 1b replicons carrying NS5B genes from genotype 2b, 5a or 6a were used for testing.

c. Data from various strains of full length NS5A replicons or chimeric NS5A replicons carrying full-length NS5A genes that contain L31 or M31 polymorphisms.

d. Data from a chimeric NS5A replicon carrying NS5A amino acids 9-184.

Table 4: Activity of sofosbuvir and velpatasvir against transient replicons containing NS5A or NS5B from clinical isolates

Replicon genotype	Replicons containing NS5B from clinical isolates	Median sofosbuvir EC ₅₀ , nM (range)	Replicons containing NS5A from clinical isolates	Median velpatasvir EC ₅₀ , nM (range)
1 ^a	67	62 (29-128)	23	0.019(0.011-0.078)
1 ^b	29	102 (45-170)	34	0.012(0.005-0.500)
2 ^a	15	29 (14-81)	8	0.011(0.006-0.364)
2 ^b	NA	NA	16	0.002(0.0003-0.307)
3 ^a	106	81 (24-181)	38	0.005(0.002-1.671)
4 ^a	NA	NA	5	0.002(0.001-0.004)
4 ^b	NA	NA	10	0.007(0.004-0.011)
4 ^c	NA	NA	7	0.003(0.002-0.006)
5 ^a	NA	NA	42	0.005(0.001-0.019)
6 ^a	NA	NA	26	0.007(0.0005-0.113)
6 ^b	NA	NA	15	0.024 (0.005-0.433)

NA = Not available

The presence of 40% human serum had no effect on the anti-HCV activity of sofosbuvir but reduced the anti-HCV activity of velpatasvir by 13-fold against genotype 1a HCV replicons.

Evaluation of sofosbuvir in combination with velpatasvir showed no antagonistic effect in reducing HCV RNA levels in replicon cells.

Resistance

In cell culture

HCV replicons with reduced susceptibility to sofosbuvir have been selected in cell culture for multiple genotypes including 1b, 2a, 2b, 3a, 4a, 5a and 6a. Reduced susceptibility to sofosbuvir was associated with the primary NS5B substitution S282T in all replicon genotypes examined. Site-directed mutagenesis of the S282T substitution in replicons of genotype 1 to b conferred 2- to 18-fold reduced susceptibility to sofosbuvir and reduced the replication viral capacity by 89% to 99% compared to the corresponding wild-type. In biochemical assays, the ability of the active triphosphate of sofosbuvir (GS-461203) to inhibit recombinant NS5B polymerase from genotypes 1b, 2a, 3a and 4a expressing the S282T substitution was reduced compared to its ability to inhibit wild-type recombinant NS5B polymerase, as indicated by a 8.5- to 24-fold increase in the 50% inhibitory concentration (IC₅₀).

fumarate, or elvitegravir/cobicistat/emtricitabine/tenofovir disoproxil fumarate Summary of Product Characteristics for recommendations on renal monitoring.

Use in diabetic patients

Diabetics may experience improved glucose control, potentially resulting in symptomatic hypoglycaemia, after initiating HCV direct-acting antiviral treatment. Glucose levels of diabetic patients initiating direct-acting antiviral therapy should be closely monitored, particularly within the first 3 months, and their diabetic medication modified when necessary. The physician in charge of the diabetic care of the patient should be informed when direct-acting antiviral therapy is initiated.

HCV/HBV (hepatitis B virus) co-infection

Cases of hepatitis B virus (HBV) reactivation, some of them fatal, have been reported during or after treatment with direct-acting antiviral agents. HBV screening should be performed in all patients before initiation of treatment. HBV/HCV co-infected patients are at risk of HBV reactivation, and should therefore be monitored and managed according to current clinical guidelines.

CPT Class C cirrhosis

Safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg has not been assessed in patients with CPT Class C cirrhosis (see sections Undesirable effects and Pharmacodynamic properties).

Liver transplant patients

The safety and efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in the treatment of HCV infection in patients who are post-liver transplant have not been assessed. Treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in accordance with the recommended posology (see section Posology) should be guided by an assessment of the potential benefits and risks for the individual patient.

4.5 Interaction with other medicinal products and other forms of interaction

As Sofosbuvir and Velpatasvir tablet contains sofosbuvir and velpatasvir, any interactions that have been identified with these active substances individually may occur with Sofosbuvir and Velpatasvir tablet.

Potential for Sofosbuvir and Velpatasvir tablet to affect other medicinal products

Velpatasvir is an inhibitor of drug transporter P-gp, breast cancer resistance protein (BCRP), organic anion-transporting polypeptide (OATP) 1B1 and OATP1B3. Co-administration of Sofosbuvir and Velpatasvir tablet with medicinal products that are substrates of these transporters may increase the exposure of such medicinal products. See Table 3 for examples of interactions with sensitive substrates of P-gp (digoxin), BCRP (rosuvastatin), and OATP (pravastatin).

Potential for other medicinal products to affect Sofosbuvir and Velpatasvir tablet

Sofosbuvir and velpatasvir are substrates of drug transporters P-gp and BCRP. Velpatasvir is also a substrate of drug transporter OATP1B. *In vitro*, slow metabolic turnover of velpatasvir by CYP2B8, CYP2C8 and CYP3A4 was observed. Medicinal products that are potent inducers of P-gp or potent inducers of CYP2B8, CYP2C8, or CYP3A4 (e.g. rifampicin, rifabutin, St. John's wort, carbamazepine, phenobarbital and phenytoin) may decrease plasma concentrations of sofosbuvir or velpatasvir leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir. The use of such medicinal products with Sofosbuvir and Velpatasvir tablet is contraindicated (see section 4.3). Medicinal products that are moderate P-gp inducers or moderate CYP inducers (e.g. oxcarbazepine, modafinil, efavirenz or rifampine) may decrease sofosbuvir or velpatasvir plasma concentration leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir tablet. Co-administration with such medicinal products is not recommended with Sofosbuvir and Velpatasvir tablet (see section 4.4). Co-administration with medicinal products that inhibit P-gp or BCRP may increase sofosbuvir or velpatasvir plasma concentrations. Medicinal products that inhibit OATP CYP2B6, CYP2C8, or CYP3A4 may increase plasma concentration of velpatasvir. Clinically significant medicinal product interactions with Sofosbuvir and Velpatasvir tablet mediated by P-gp, BCRP, OATP or CYP450 inhibitors are not expected. Sofosbuvir and Velpatasvir tablet may be co-administered with P-gp, BCRP, OATP and CYP inhibitors.

Patients treated with vitamin K antagonists

As liver function may change during treatment with Sofosbuvir and Velpatasvir, a close monitoring of International Normalised Ratio (INR) values is recommended.

Impact of DAA therapy on drugs metabolized by the liver:

The pharmacokinetics of drugs that are metabolized by the liver (e.g. immunosuppressive agents such as calcineurin inhibitors) may be impacted by changes in liver function during DAA therapy, related to clearance of HCV.

Interactions between Sofosbuvir and Velpatasvir tablet and other medicinal products

Table 3 provides a listing of established or potentially clinically significant medicinal product interactions. The medicinal product interactions described are based on studies conducted with either Sofosbuvir and Velpatasvir or velpatasvir and sofosbuvir as individual agents, or are predicted medicinal product interactions that may occur with Sofosbuvir OR Velpatasvir. The table is not all-inclusive.

Table 3: Interactions between Sofosbuvir and Velpatasvir tablet and other medicinal products

Medicinal product by therapeutic areas/Possible Mechanism of Interaction	Recommendation concerning co-administration with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg
ACID REDUCING AGENTS	
	Velpatasvir solubility decreases as pH increases. Medicinal products that increase gastric pH are expected to decrease the concentration of velpatasvir.
Antacids	
e.g. Aluminium or magnesium hydroxide; calcium carbonate (Increase in gastric pH)	It is recommended to separate antacid and Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg administration by 4 hours.
H₂-receptor antagonists	
Famotidine (40 mg single dose)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose) ^a Famotidine dosed simultaneously with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a Cimetidine ^a Nizatidine ^a Ranitidine ^a (Increase in gastric pH)	H ₂ -receptor antagonists may be administered simultaneously with or staggered from Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg at a dose that does not exceed doses comparable to famotidine 40 mg twice daily.
Proton pump inhibitors	
Famotidine (40 mg single dose)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose) ^a Famotidine dosed 12 hours prior to Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a (Increase in gastric pH)	
Proton pump inhibitors	
Omeprazole (20 mg once daily)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose fasted) ^a Omeprazole dosed simultaneously with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a Lansoprazole ^a Rabeprazole ^a Pantoprazole ^a Esomeprazole ^a (Increase in gastric pH)	Co-administration with proton pump inhibitors is not recommended. If it is considered necessary to co-administer, then Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg should be administered with food and taken at least 2 hours before the proton pump inhibitor at max doses comparable to omeprazole 20 mg.
Omeprazole (20 mg once daily)/ sofosbuvir/ velpatasvir (400/ 100 mg single dose fed) ^a Omeprazole dosed 4 hours after Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg ^a (Increase in gastric pH)	
ANTIARRHYTHMICS	
Amiodarone	Use only if no other alternative is available. Close monitoring is recommended if this medicinal product is administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg (see sections 4.4 and 4.8).
Digoxin Digoxin (0.25 mg single dose)/ velpatasvir (100 mg single dose) (Inhibition of P-gp)	Co-administration of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg with digoxin may increase the concentration of digoxin. Caution is warranted and therapeutic concentration monitoring of digoxin is recommended when co-administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.
ANTICOAGULANTS	
Dabigatran etexilate (Inhibition of P-gp)	Clinical monitoring, looking for signs of bleeding and anaemia, is recommended when dabigatran etexilate is co-administered with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. A coagulation test helps to identify patients with an increased bleeding risk due to increased dabigatran exposure.
Vitamin K antagonists	Close monitoring of INR is recommended with all vitamin K antagonists. This is due to liver function changes during treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.
ANTICONSULTANTS	
Carbamazepine Phenytoin Phenobarbital (Induction of P-gp and CYPs)	Sofosbuvir and Velpatasvir Film Coated tablets 400 mg/100 mg is contraindicated with carbamazepine, phenobarbital and phenytoin, potent P-gp and CYP inducers (see section 4.3).
Oxcarbazepine (Induction of P-gp and CYPs)	Co-administration of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg with oxcarbazepine is expected to decrease the concentration of sofosbuvir and velpatasvir, leading to reduced therapeutic effect of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg. Co-administration is not recommended (see section 4.4).

In vitro selection of HCV replicons with reduced susceptibility to velpatasvir was performed in cell culture for multiple genotypes including 1a, 1b, 2a, 3a, 4a, 5a and 6a. Variants were selected at NS5A resistance associated positions 24, 28, 30, 31, 32, 58, 92 and 93. The resistance associated variants (RAVs) selected in 2 or more genotypes were F28S, L31V and Y93H. Site-directed mutagenesis of known NS5A RAVs showed that substitutions conferring a > 100-fold reduction in velpatasvir susceptibility are M28G, A92K and Y93H N/R/W in genotype 1a, A92K in genotype 1b, C92T and Y93H in genotype 2b, Y93H in genotype 3, and L31V and P32A/LQR in genotype 6. No individual substitutions tested in genotypes 2a, 4a, or 5a conferred a > 100-fold reduction in velpatasvir susceptibility. Combinations of these variants often showed greater reductions in susceptibility to velpatasvir than single RAVs alone.

In clinical studies Studies in patients without cirrhosis and patients with compensated cirrhosis In a pooled analysis of patients without cirrhosis or with compensated cirrhosis who received Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks in three Phase 3 studies, 12 patients (2 with genotype 1 and 10 with genotype 3) qualified for resistance analysis due to virologic failure. One additional patient with genotype 3 HCV infection at baseline was reinfected with genotype 1a HCV at virologic failure and was excluded from the virological analysis. No patients with genotype 2, 4, 5, or 6 HCV infection experienced virologic failure.

Of the 2 genotype 1 virologic failure patients, one patient had virus with emergent NS5A RAVs L31V and Y93H at virologic failure. Both patients had virus at baseline harboring NS5A RAVs. No NS5B nucleoside inhibitor (NI) RAVs were observed at failure in the 2 patients. Of the 10 genotype 3 virologic failure patients, Y93H was observed in all 10 patients at failure (6 had Y93H emerge post-treatment and 4 patients had Y93H at baseline and post-treatment). No NS5B NI RAVs were observed at failure in the 10 patients.

Studies in patients with decompensated cirrhosis

In one Phase 3 study in patients with decompensated cirrhosis who received Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV for 12 weeks, 3 patients (1 with genotype 1 and 2 with genotype 3) qualified for resistance analysis due to virologic failure. No patients with genotype 2 or 4 HCV infection in the Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 weeks group experienced virologic failure. The 1 virologic failure patient with genotype 1 HCV had no NS5A or NS5B RAVs at failure. Of the 2 genotype 3 virologic failure patients, one had NS5A RAV Y93H emerge at failure. Another patient had virus with Y93H at baseline and virologic failure and also developed low levels (< 5%) of NS5B NI RAVs N142T and E237G at failure. Pharmacokinetic data from this patient was consistent with non-adherence to treatment.

In this study, 2 patients treated with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 or 24 weeks without ribavirin had emergent NS5B S282T at low levels (< 5%) failure with L159F.

Effect of baseline HCV resistance-associated variants on treatment outcome

Studies in patients without cirrhosis and patients with compensated cirrhosis

Analyses were conducted to explore the association between pre-existing baseline NS5A RAVs and treatment outcome for patients without cirrhosis or with compensated cirrhosis in three Phase 3 clinical studies (ASTRAL-1, ASTRAL-2 and ASTRAL-3). Of the 1,035 patients treated with Sofosbuvir and Velpatasvir in the three Phase 3 clinical studies, 1,023 patients were included in the analysis of NS5A RAVs; 7 patients were excluded as they neither achieved sustained virologic response (SVR12) nor had virologic failure and 5 additional patients were excluded as NS5A gene sequencing failed. In the pooled analysis of the Phase 3 studies, 3801/023 (37%) patients' virus had baseline NS5A RAVs. Genotype 2, 4, and 6 HCV-infected patients had a higher prevalence of NS5A RAVs (70%, 63% and 52%, respectively) compared to genotype 1 (23%), genotype 3 (16%), and genotype 5 (18%) HCV-infected patients.

Baseline RAVs had no relevant impact on SVR12 rates in patients infected with genotype 1, 2, 4, 5 and 6 HCV as summarised in Table 5. Genotype 3 infected patients with the NS5A RAV Y93H at baseline had a lower SVR12 rate than patients without Y93H after treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks, as summarised in Table 6. In the ASTRAL-3 study, the Y93H RAV was detected at baseline in 9% of patients treated with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg.

Table 5: SVR12 in patients with or without baseline NS5A RAVs by HCV genotype (studies ASTRAL-1, ASTRAL-2 and ASTRAL-3)

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks				
	Genotype 1	Genotype 3	Genotypes 2, 4, 5 or 6	Total	
With any baseline NS5A RAVs	97% (73/75)	88% (38/43)	100% (262/262)	98% (373/390)	
Without baseline NS5A RAVs	100% (251/251)	97% (225/231)	100% (161/161)	99% (637/643)	

Table6: SVR12 in patients with and without baseline Y93H, 1% Cut-off (Resistance Analysis Population Set) ASTRAL-3

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 Weeks			
	All Subjects (n=274)	Cirrhotic (n=80)	Non-Cirrhotic (n=197)	
Overall	95.3% (263/274)	91.3% (73/80)	97.9% (190/194)	
95% CI	92.9% to 98.0%	82.8% to 96.4%	92.8% to 98.6%	
SVR with Y93H	84.0% (11/25)	50.0% (2/4)	90.5% (19/21)	
95% CI	63.9% to 95.5%	6.8% to 93.2%	69.6% to 98.8%	
SVR without Y93H	96.4% (242/249)	93.4% (71/76)	98.8% (171/173)	
95% CI	94.3% to 98.9%	85.3% to 99.8%	95.9% to 99.9%	

The NS5B NI RAV S282T was not detected in the baseline NS5B sequence of any patient in Phase 3 studies. SVR12 was achieved in all 77 patients who had baseline NS5B NI RAVs including N142T, L159F, E2N37G, C/M289L1, L320F/IV, V321A/I, and S282G + V321L.

Studies in patients with decompensated cirrhosis (CPT Class B)

Analyses were conducted to explore the association between pre-existing baseline NS5A RAVs and treatment outcome for patients with decompensated cirrhosis in one Phase 3 study (ASTRAL-4). Of the 87 patients treated with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV, 65 patients were included in the analysis of NS5A RAVs; 2 patients were excluded as they neither achieved SVR12 nor had virologic failure. Among the patients who received treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV for 12 weeks, 29% (25/65) of patients had baseline virologic NS5A RAVs: 29% (19/66), 75% (3/4), 15% (2/13), and 50% (1/2) for patients with genotype 1, 2, 3 and 4 HCV, respectively.

SVR12 in patients with or without baseline NS5A RAVs in the Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 week group for this study is shown in Table 7.

Table 7: SVR12 in patients with or without baseline NS5A RAVs by HCV genotype (study ASTRAL-4)

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 weeks			
	Genotype 1	Genotype 3	Genotypes 2 or 4	Total
With any baseline NS5A RAVs	100% (19/19)	50% (1/2)	100% (4/4)	96% (24/25)
Without baseline NS5A RAVs	98% (46/47)	91% (10/11)	100% (2/2)	98% (58/60)

The single genotype 3 patient who had baseline NS5A RAVs and failed to achieve SVR12 had NS5A substitution Y93H at baseline; pharmacokinetic data from this patient was consistent with non-adherence to treatment.

Three patients in the Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 week group had baseline NS5B NI RAVs (N142T and L159F) and all three patients achieved SVR12.

Cross-resistance

In vitro data suggests that the majority of NS5A RAVs that confer resistance to ledipasvir and daclatasvir remained susceptible to velpatasvir. Velpatasvir was fully active against the sofosbuvir resistance-associated substitution S282T in NS5B while all velpatasvir resistance-associated substitutions in NS5A were fully susceptible to sofosbuvir. Both sofosbuvir and velpatasvir were fully active against substitutions associated with resistance to other classes of direct acting antivirals with different mechanisms of actions, such as NS5B non-nucleoside inhibitors and NS3 protease inhibitors. The efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg has not been assessed in patients who have previously failed treatment with other regimens that include an NS5A inhibitor.

Clinical efficacy and safety

The efficacy of Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg was evaluated in three Phase 3 studies in patients with genotype 1 to 6 HCV infection with or without compensated cirrhosis one Phase 3 study in patients with genotype 1 to 6 HCV infection with decompensated cirrhosis, and one Phase 3 study in HCV/HIV-1 co-infected patients with genotype 1 to 6 HCV infection, as summarised in Table 8.

Table 8: Studies conducted with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in patients with genotype 1, 2, 3, 4, 5 or 6 HCV infection

Study	Population	Study arms (Number of patients treated)
ASTRAL-1	Genotype 1, 2, 4, 5 and 6 TN and TE, without cirrhosis or with compensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (624)
	Genotype 2 TN and TE, without cirrhosis or with compensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (134) SOF+RBV 12 weeks (132)
ASTRAL-2	Genotype 3 TN and TE, without cirrhosis or with compensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (277)
	Genotype 1, 2, 3, 4, 5 and 6 TN and TE, with CPT Class B decompensated cirrhosis	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (90) Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 weeks (87) Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 24 weeks (90)
ASTRAL-3	Genotype 1, 2, 3, 4, 5 and 6 TN and TE, without cirrhosis or with compensated cirrhosis, with HCV/HIV-1 co-infection	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (106)

TN = treatment-naïve patients; TE = treatment-experienced patients (including those who have failed a peginterferon alpha + ribavirin based regimen with or without an HCV protease inhibitor)

The ribavirin dose was weight-based (1,000 mg daily administered in two divided doses for patients < 75 kg and 1,200 mg for those ≥ 75 kg) and administered in two divided doses when used in combination with sofosbuvir in the ASTRAL-2 and ASTRAL-3 studies or in combination with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg in the ASTRAL-4 study. Ribavirin dose adjustments were performed according to the ribavirin prescribing information. Serum HCV RNA values were measured during the clinical studies using the COBAS AmpliPrep/COBAS Taqman HCV test (version 2.0) with a lower limit of quantification (LLOQ) of 15 IU/mL. Sustained virologic response (SVR12), defined as HCV RNA less than LLOQ at 12 weeks after the cessation of treatment, was the primary endpoint to determine the HCV cure rate.

Clinical studies in patients without cirrhosis and patients with compensated cirrhosis Genotype 1, 2, 4, 5 and 6 HCV-infected adults – ASTRAL-1 (study 1138)

ASTRAL-1 was a randomised, double-blind, placebo-controlled study that evaluated 12 weeks of treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg compared with 12 weeks of placebo in patients with genotype 1, 2, 4, 5, or 6 HCV infection. Patients with genotype 1, 2, 4, or 6 HCV infection were randomised in a 5:1

ratio to treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks or placebo for 12 weeks. Patients with genotype 5 HCV infection were enrolled in the Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg group. Randomisation was stratified by HCV genotype (1, 2, 4, 6, and indeterminate) and the presence or absence of cirrhosis.

Demographics and baseline characteristics were balanced between the Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg and placebo group. Of the 740 treated patients, the median age was 56 years (range: 18 to 82); 60% of the patients were male; 79% were White, 9% were Black; 21% had a baseline body mass index of at least 30 kg/m²; the proportions of patients with genotype 1, 2, 4, 5, or 6 HCV infection were 53%, 17%, 19%, 5% and 7%, respectively; 69% had non-CC IL28B alleles (CT or TT); 74% had baseline HCV RNA levels of at least 800,000 IU/mL; 19% had compensated cirrhosis, and 32% were treatment-experienced.

Table 9 presents the SVR12 for the ASTRAL-1 study by HCV genotypes. No patients in the placebo group achieved SVR12.

Table 9: SVR12 in study ASTRAL-1 by HCV genotype

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (n = 624)					
	Total (all Gts) (n=624)	GT-1a (n=210)	GT-1b (n=118)	Total (n=328)	GT-2 (n=104)	GT-4 (n=116)
SVR12	99% (618/624)	98% (206/210)	91% (117/118)	98% (323/328)	100% (104/104)	97% (116/116)
Outcome for patients without SVR12						
On-treatment virologic failure	0/624	0/210	0/118	0/328	0/104	0/116
Relapse ^a	< 1% (2/623)	< 1% (1/209)	1% (1/118)	2% (2/327)	0/104	0/116
Other ^a	1% (1/624)	1% (3/210)	0/118	1% (3/328)	0/104	0/116

GT = genotype

- The denominator for relapse is the number of patients with HCV RNA < LLOQ at their last on-treatment assessment.
- Other includes patients who did not achieve SVR12 and did not meet virologic failure criteria.

Genotype 2 HCV-infected adults – ASTRAL-2 (study 1139)

ASTRAL-2 was a randomised, open-label study that evaluated 12 weeks of treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg compared with 12 weeks of treatment with SOF+RBV in patients with genotype 2 HCV infection. Patients were randomised in a 1:1 ratio to treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks or SOF+RBV for 12 weeks. Randomisation was stratified by the presence or absence of cirrhosis and prior treatment experience (treatment-naïve versus treatment-experienced).

Demographics and baseline characteristics were balanced across the two treatment groups. Of the 266 treated patients, the median age was 58 years (range: 23 to 81); 59% of the patients were male; 88% were White, 7% were Black; 33% had a baseline body mass index of at least 30 kg/m²; 62% had non-CC IL28B alleles (CT or TT); 80% had baseline HCV RNA levels of at least 800,000 IU/mL; 14% had compensated cirrhosis and 15% were treatment-experienced.

Table 10 presents the SVR12 for the ASTRAL-2 study.

Table 10: SVR12 in study ASTRAL-2 (HCV genotype 2)

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (n = 134)	SOF + RBV12 weeks (n = 132)
SVR12	99% (133/134)	94% (124/132)
Outcome for patients without SVR12		
On-treatment virologic failure	0/134	0/132
Relapse ^a	0/133	5% (6/132)
Other ^a	1% (1/134)	2% (2/132)

- The denominator for relapse is the number of patients with HCV RNA < LLOQ at their last on-treatment assessment.
- Other includes patients who did not achieve SVR12 and did not meet virologic failure criteria.

Treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks demonstrated the statistical superiority (p = 0.018) over treatment with SOF+RBV for 12 weeks (treatment difference +5.2%; 95% confidence interval: +0.2% to +10.3%).

Genotype 3 HCV-infected adults – ASTRAL-3 (study 1140)

ASTRAL-3 was a randomised, open-label study that evaluated 12 weeks of treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg compared with 24 weeks of treatment with SOF+RBV in patients with genotype 3 HCV infection. Patients were randomised in a 1:1 ratio to treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks or SOF+RBV for 24 weeks. Randomisation was stratified by the presence or absence of cirrhosis and prior treatment experience (treatment-naïve versus treatment-experienced).

Demographics and baseline characteristics were balanced across the two treatment groups. Of the 552 treated patients, the median age was 52 years (range: 19 to 76); 62% of the patients were male; 89% were White, 9% were Asian; 1% were Black; 20% had a baseline body mass index of at least 30 kg/m²; 61% had non-CC IL28B alleles (CT or TT); 70% had baseline HCV RNA levels of at least 800,000 IU/mL; 30% had compensated cirrhosis and 26% were treatment-experienced.

Table 11 presents the SVR12 for the ASTRAL-3 study.

Table 11: SVR12 in study ASTRAL-3 (HCV genotype 3)

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (n = 277)	SOF + RBV 24 weeks (n = 275)
SVR12	95% (264/277)	80% (221/275)
Outcome for patients without SVR12		
On-treatment virologic failure	0/277	< 1% (1/275)
Relapse ^a	4% (11/276)	14% (38/272)
Other ^a	1% (2/277)	5% (15/275)

- The denominator for relapse is the number of patients with HCV RNA < LLOQ at their last on-treatment assessment.
- Other includes patients who did not achieve SVR12 and did not meet virologic failure criteria.

Treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks demonstrated the statistical superiority (p < 0.001) compared to treatment with SOF+RBV for 24 weeks (treatment difference +14.8%; 95% confidence interval: +9.6% to +20.0%).

SVR12 for selected subgroups are presented in Table 12.

Table 12: SVR12 for selected subgroups in study ASTRAL-3 (HCV genotype 3)

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks		SOF+RBV 24 weeks ^a	
SVR12	Treatment-naïve (n = 208)	Treatment-experienced (n = 201)	Treatment-naïve (n = 69)	Treatment-experienced (n = 69)
Without cirrhosis	98% (190/163)	91% (31/34)	90% (141/156)	71% (22/31)
With cirrhosis	93% (40/43)	89% (33/37)	73% (33/45)	58% (22/38)

- Five patients with missing cirrhosis status in the SOF+RBV 24 week group were excluded from this subgroup analysis.

Clinical studies in patients with decompensated cirrhosis – ASTRAL-4 (study 1137)

ASTRAL-4 was a randomised, open-label study in patients with genotype 1, 2, 3, 4, 5 or 6 HCV infection and CPT Class B cirrhosis. Patients were randomised in a 1:1:1 ratio to treatment with Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 12 weeks, Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV for 12 weeks or Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg for 24 weeks. Randomisation was stratified by HCV genotype (1, 2, 3, 4, 5, 6 and indeterminate).

Demographics and baseline characteristics were balanced across the treatment groups. Of the 267 treated patients, the median age was 59 years (range: 40 to 73); 70% of the patients were male; 90% were White, 6% were Black; 42% had a baseline body mass index of at least 30 kg/m². The proportions of patients with genotype 1, 2, 3, 4 or 6 HCV were 78%, 4%, 15%, 3%, and < 1% (1 patient), respectively. No patients with genotype 5 HCV infection were enrolled. 78% of the patients had non-CC IL28B alleles (CT or TT); 56% had baseline HCV RNA levels of at least 800,000 IU/mL; 55% were treatment-experienced; 90% and 95% of patients had CPT Class B cirrhosis and Model for End Stage Liver Disease (MELD) score ≤ 15 at baseline, respectively.

Table 13 presents the SVR12 for the ASTRAL-4 study by HCV genotype.

Table 13: SVR12 in study ASTRAL-4 by HCV genotype

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks (n = 90)		Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 weeks (n = 87)	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 24 weeks (n = 90)
Overall SVR12	83% (75/90)		94% (82/87)	86% (77/90)
Genotype 1 ^a	88% (60/68)		96% (65/68)	92% (65/71)
Genotype 1 ^b	89% (16/18)		100% (14/14)	88% (14/16)
Genotype 3	50% (7/14)		85% (11/13)	50% (6/12)
Genotype 2, 4 and 6	100% (8/8)		100% (6/6)	86% (6/7)

- n = 4 for genotype 2 and n = 4 for genotype 4
- n = 4 for genotype 2 and n = 2 for genotype 4
- n = 4 for genotype 2, n = 2 for genotype 4 and n = 1 for genotype 6

Table 14 presents the virologic outcome for patients with genotype 1 or 3 HCV infection in the ASTRAL-4 study. No patients with genotype 2, 4 or 6 HCV infection experienced virologic failure.

Table 14: Virologic outcome for patients with genotype 1 and 3 HCV infection in study ASTRAL-4

	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 12 weeks	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg + RBV 12 weeks	Sofosbuvir and Velpatasvir Film Coated Tablets 400 mg/100 mg 24 weeks
Virologic failure (relapse and on-treatment failure)			
Genotype 1 ^a	7% (5/68)	1% (1/68)	4% (3/71)
Genotype 1 ^b	6% (3/50)	2% (1/54)	4% (2/55)
Genotype 1 ^c	11% (2/18)	0% (0/14)	6% (1/16)
Genotype 3	43% (6/14)	15% (2/13)	42% (5/12)
Other ^a	5% (4/82)	2% (2/81)	5% (4/83)

- No patients with genotype 1 HCV had on-treatment virologic failure.
- One patient had on-treatment virologic failure; pharmacokinetic data from this patient was consistent with non-adherence to treatment.
- One patient had on-treatment virologic failure.
- Other includes patients who did not achieve SVR12 and did not meet virologic failure criteria.

Changes in the parameters found in the CPT score system in patients achieving SVR12 in ASTRAL-4 (all 3 regimens) are shown in Table 15.

Table 15: Changes in CPT score parameters from baseline to week 12 and 24 post-treatment in patients achieving SVR12, ASTRAL-4

	Albumin	Bilirubin	INR	Ascites	Encephalopathy
Post-treatment Week 12 (N=236), % (n/N)					
Decreased score (Improvement)	34.5% (79/229)	17.9% (41/229)	2.2% (5/229)	7.9% (18/229)	5.2% (12/229)
No change	60.3% (138/229)	76.4% (175/229)	96.5% (221/229)	89.1% (204/229)	91.3% (209/229)
Increased score (Worsening)	5.2% (12/229)	5.7% (13/229)	1.3% (3/229)	3.1% (7/229)	3.5% (8/229)
No assessment	7	7	7	7	7
Post-treatment Week 24 (N=236), % (n/N)					
Decreased score (Improvement)	39.4% (84/213)	16.4% (35/213)	2.3% (5/213)	15.0% (32/213)	9.4% (20/213)
No change	54.0% (115/213)	80.8% (172/213)	94.8% (202/213)	81.2% (173/213)	88.3% (188/213)
Increased score (Worsening)	6.6% (14/213)	2.8% (6/213)	2.3% (5/213)	3.8% (8/213)	2.3% (5/213)
No assessment	23	23	23	23	23